Question			Answer	Marks	AO	Guidance
12	(a)		$\frac{3a-x}{a+x} = \frac{3a}{a} = 3 \geq 0$	**B1** **[1]**	**2.1**	• AG • Or complete argument e.g. $a > 0$ so $\frac{3a}{a} \geq 0$ • Condone $3 > 0$ or $\frac{3a}{a} > 0$ • No wrong working seen
12	(b)		$y^2 \geq 0$ and $x^2 > 0$ Since $y^2 = \frac{x^2(3a-x)}{a+x}$ then $\frac{(3a-x)}{a+x} \geq 0$	**B1** **[1]**	**2.1**	• AG • Clear explanation, must be based on y^2 non-negative (≥ 0) and x^2 positive (> 0) • Condone $\frac{y^2}{x^2} \geq 0$ then $\frac{(3a-x)}{a+x} \geq 0$ from rearranging
12	(c)		$x + a$ cannot be 0 [as then the value of y^2 is undefined]	**B1**	**2.2a**	• Dismissing zero value for $x + a$ • Allow e.g. $x + a > 0$ **and** $x + a < 0$ • Condone e.g. " $x + a > 0$ as denominator cannot be 0"
			$3a - x \leq 0$ and $a + x < 0$ ($3a - x \geq 0$ and $a + x > 0$)	**B1**	**2.2a**	• Condone "both positive or both negative" • Considering negative case • Condone e.g. $3a - x < 0$ and $a + x < 0$ for this mark
			If $3a - x \leq 0$ and $a + x < 0$ then $3a \leq x$ and $x < -a$ These cannot both be true when a is positive so $3a - x \geq 0$ and $a + x > 0$	**B1**	**2.3**	• AG • Dismissing the "both negative" case • Dep on previous two B1 marks • Inequality signs must be correct
			Alternative method for 3rd B1 If $3a - x \leq 0$ and $a + x < 0$ then $3a - x + a + x < 0 \Rightarrow 4a < 0$ but this is impossible as $a > 0$ so $3a - x \geq 0$ and $a + x > 0$	**B1**		• AG • Dismissing the "both negative" case • Dep on previous two B1 marks • Inequality signs must be correct
				[3]		

Question			Answer	Marks	AO	Guidance
12	(d)		$3a \geq x$ and $x > -a$ OR $-a < x \leq 3a$	**B1** **[1]**	**3.1a**	• Any equivalent inequality/inequalities with x as a subject • Both inequality signs must be correct • Condone e.g. comma for 'and'
12	(e)		(Asymptote is) $x = -a$	**B1** **[1]**	**2.2a**	• $-a$ alone is insufficient. • oe e.g. $x + a = 0$